

Notes from the meeting I had with Tom and his opinions on my internship tasks:

Eating and Rumination minutes:

- Point of data availability, the sensors that capture the data isn't available for all farms. So it is recommended to make those data points an extra bonus feature to support the analysis and model.
- **For modeling** it might be good to compare models using those features and ones not.
- Also recommended to use walking minutes instead of eating/rumination minutes for my case.
- **Question?** Can the sensor distinguish the difference between a cow that is ruminating and walking at the same time.

Milk settings optimizer:

- Divide it into two parts
 - Animal Settings
 - Device/Technical settings:
- What are logical numbers?
 - Late lactation cows are different from first lactation cows.
 - What is usually done is the first and second lactation cows are grouped together and then there's 3+ lactation cows.
- Lactation number group (are there different settings for them)
 - Lactation 1 is set differently for Heifers
 - Look through the data on Databricks, there are some group settings there that can be traced back to the animal with regard to milk access.
 - **There is some variation in this question:**
 - Like sometimes the farmer will put a set of cows to dry one day early because he has to do three others and do them in one go
 - This is a good one to check to really see how proactive farmers are with this
 - **Recommended** to try and find documentation on why this is being done.
Example: A cow is being put 10 days earlier than you would expect, is it because of health issues? What was in the last milking in there?
- Number of groups switches in the last 14 days:
 - That will almost never happen. Don't expect too much from this.
 - Being put into Lactation groups means you want each cow to be treated the same and have the same settings, that how it is setup in farm. Nothing is really finetuned.
 - **An example of what they start doing is:** If the cow is expected to enter dry off at 300 days in lactation, 30 days before those 300 days they will start decreasing the target

milk yield, like decreasing the amount of visits you can do to prepare for dry off. (found on the animal settings)

Are there differences in settings within a farm or more between farms?

- In regards to different lactation groups there is.
Milk under supervision:
- Mostly for cows that are problematic.
- Usually happens in cases where for example a cow fails milking 5 times in a day and they don't know why.
- **Questions to explore and look at:** For example look at does the avg count per day increase as days in lactation increases. (maybe the udder is irritated or infected, how does it proceed)
- Not expected to get many results from this, also because it is up to the farmer to decide when it's valid to do it. So there's no filter in between, some farmers are proactive with it some are not.

For phase 2:

What parameters impact dry-off:

- Lactation number
- **Productivity class:** Do you have cows that are high yield relative to those that are lower production level.
- **For country:** More of a correction factor than a determinant factor.
 - **Start simple:** Take a few Dutch farms and because those are roughly comparable with their management strategies and then start extrapolating this to northwestern Europe.
 - And then really start looking at, the zones with really some differences in climate. And then think about Scandinavia, Southern Europe, Australia, because these are settings in which the production systems are not only the climate is very different also production settings.
 - So start small with Dutch farms, we know that year round the data from Dutch farms is most reliable.